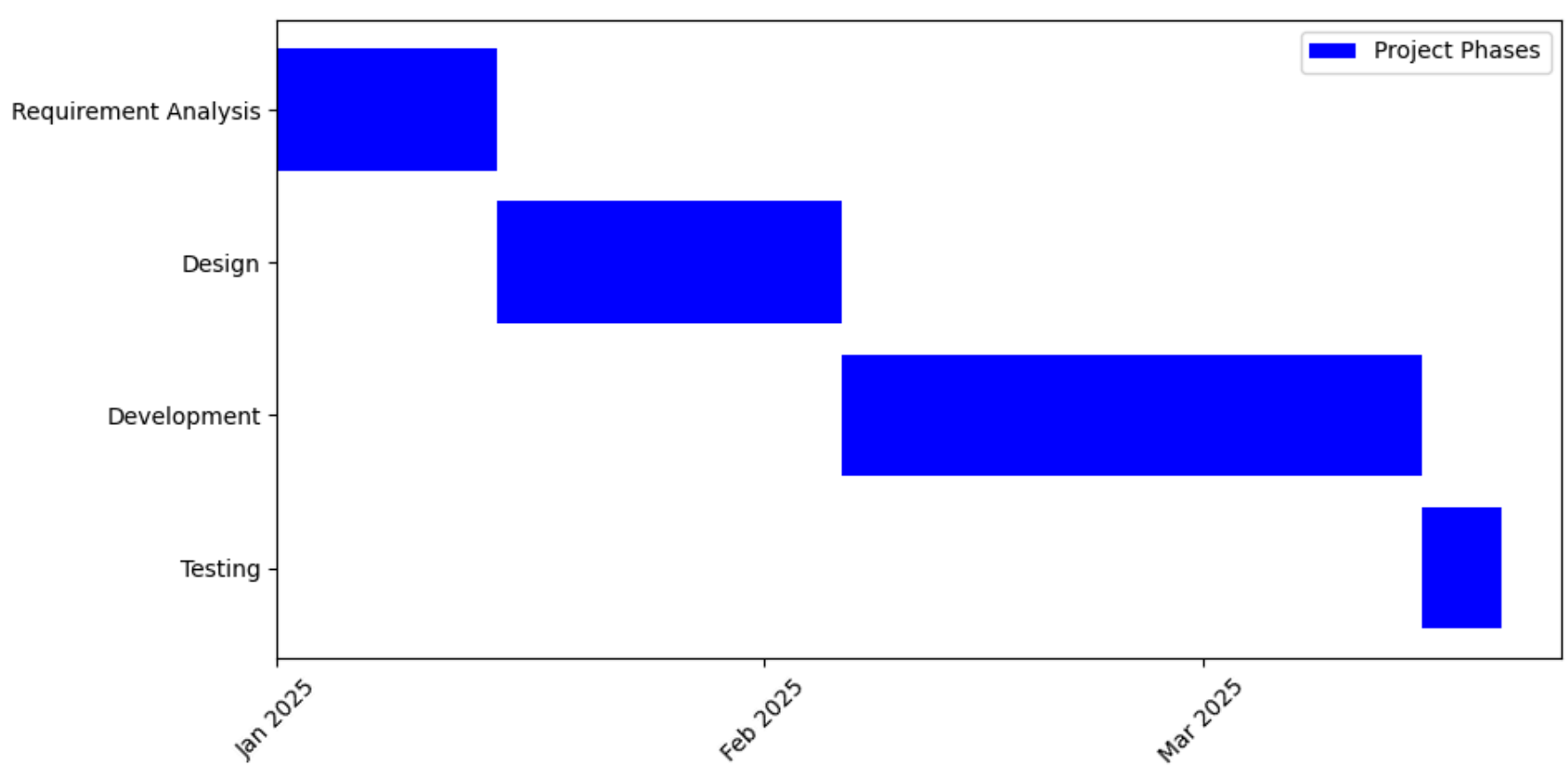
Gantt chart

```
import matplotlib.pyplot as plt
import pandas as pd
import matplotlib.dates as mdates

# Define project phases and their timelines
phases = ["Requirement Analysis", "Design", "Development", "Testing"]
start_dates = ["2025-01-01", "2025-01-15", "2025-02-06", "2025-03-15"]
end_dates = ["2025-01-15", "2025-02-06", "2025-03-15", "2025-03-20"]

# Convert dates to pandas datetime format
start_dates = pd.to_datetime(start_dates)
end_dates = pd.to_datetime(end_dates)
durations = end_dates - start_dates

# Create figure and axis
fig, ax = plt.subplots(figsize=(10, 5))

# Plot bars for each phase (reversed order of y-axis)
for i, (phase, start, duration) in enumerate(zip(reversed(phases), reversed(start_dates), reversed(durations))) :
    ax.barh(phase, duration.days, left=start, color='blue', label="Project Phases" if i == 0 else "")

# Formatting x-axis labels
ax.xaxis.set_major_locator(mdates.MonthLocator())
ax.xaxis.set_major_formatter(mdates.DateFormatter('%b %Y'))
plt.xticks(rotation=45)

# Labels and title

# Legend
ax.legend()

# Save the chart
gantt_chart_reversed_path = "/mnt/data/gantt_chart_reversed.png"
#plt.savefig(gantt_chart_reversed_path)
plt.show()

# Provide download link
gantt_chart_reversed_path
```




```
'/mnt/data/gantt_chart_reversed.png'
```

Installing and importing libraries

```
!pip install matplotlib
!pip install numpy
!pip install opencv-python
!pip install os
```

```
!pip install tensorflow
!pip install tqdm
```

```
Requirement already satisfied: matplotlib in /usr/local/lib/python3.11/dist-packages (3.10.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.11/dist-packages (from matplotlib) (1.3.1)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.11/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.11/dist-packages (from matplotlib) (4.57.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.11/dist-packages (from matplotlib) (1.4.8)
Requirement already satisfied: numpy>=1.23 in /usr/local/lib/python3.11/dist-packages (from matplotlib) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.11/dist-packages (from matplotlib) (24.2)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.11/dist-packages (from matplotlib) (11.1.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.11/dist-packages (from matplotlib) (3.2.3)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.11/dist-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
Requirement already satisfied: numpy in /usr/local/lib/python3.11/dist-packages (2.0.2)
Collecting opencv-python
  Downloading opencv_python-4.11.0.86-cp37-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (20 kB)
Requirement already satisfied: numpy>=1.21.2 in /usr/local/lib/python3.11/dist-packages (from opencv-python) (2.0.2)
Downloading opencv_python-4.11.0.86-cp37-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (63.0 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 63.0/63.0 MB 20.6 MB/s eta 0:00:00
Installing collected packages: opencv-python
Successfully installed opencv-python-4.11.0.86
ERROR: Could not find a version that satisfies the requirement os (from versions: none)
ERROR: No matching distribution found for os
Collecting tensorflow
  Downloading tensorflow-2.19.0-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (4.1 kB)
Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.11/dist-packages (from tensorflow) (1.4.0)
Collecting astunparse>=1.6.0 (from tensorflow)
  Downloading astunparse-1.6.3-py2.py3-none-any.whl.metadata (4.4 kB)
Collecting flatbuffers>=24.3.25 (from tensorflow)
  Downloading flatbuffers-25.2.10-py2.py3-none-any.whl.metadata (875 bytes)
Requirement already satisfied: gast!=0.5.0,!=0.5.1,!=0.5.2,>=0.2.1 in /usr/local/lib/python3.11/dist-packages (from tensorflow) (0.6.0)
Collecting google-pasta>=0.1.1 (from tensorflow)
  Downloading google_pasta-0.2.0-py3-none-any.whl.metadata (814 bytes)
Collecting libclang>=13.0.0 (from tensorflow)
  Downloading libclang-18.1.1-py2.py3-none-manylinux2010_x86_64.whl.metadata (5.2 kB)
Requirement already satisfied: opt-einsum>=2.3.2 in /usr/local/lib/python3.11/dist-packages (from tensorflow) (3.4.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.11/dist-packages (from tensorflow) (24.2)
Requirement already satisfied: protobuf!=4.21.0,!=4.21.1,!=4.21.2,!=4.21.3,!=4.21.4,!=4.21.5,<6.0.0dev,>=3.20.3 in /usr/local/lib/pytho
Requirement already satisfied: requests<3,>=2.21.0 in /usr/local/lib/python3.11/dist-packages (from tensorflow) (2.32.3)
Requirement already satisfied: setuptools in /usr/local/lib/python3.11/dist-packages (from tensorflow) (75.2.0)
Requirement already satisfied: six>=1.12.0 in /usr/local/lib/python3.11/dist-packages (from tensorflow) (1.17.0)
Requirement already satisfied: termcolor>=1.1.0 in /usr/local/lib/python3.11/dist-packages (from tensorflow) (3.0.1)
Requirement already satisfied: typing-extensions>=3.6.6 in /usr/local/lib/python3.11/dist-packages (from tensorflow) (4.13.1)
Requirement already satisfied: wrapt>=1.11.0 in /usr/local/lib/python3.11/dist-packages (from tensorflow) (1.17.2)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in /usr/local/lib/python3.11/dist-packages (from tensorflow) (1.71.0)
Collecting tensorboard~=2.19.0 (from tensorflow)
  Downloading tensorboard-2.19.0-py3-none-any.whl.metadata (1.8 kB)
Requirement already satisfied: keras>=3.5.0 in /usr/local/lib/python3.11/dist-packages (from tensorflow) (3.8.0)
Requirement already satisfied: numpy<2.2.0,>=1.26.0 in /usr/local/lib/python3.11/dist-packages (from tensorflow) (2.0.2)
Requirement already satisfied: h5py>=3.11.0 in /usr/local/lib/python3.11/dist-packages (from tensorflow) (3.13.0)
Requirement already satisfied: ml-dtypes<1.0.0,>=0.5.1 in /usr/local/lib/python3.11/dist-packages (from tensorflow) (0.5.1)
Collecting tensorflow-io-gcs-filesystem>=0.23.1 (from tensorflow)
  Downloading tensorflow_io_gcs_filesystem-0.37.1-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (14 kB)
Collecting wheel<1.0,>=0.23.0 (from astunparse>=1.6.0->tensorflow)
  Downloading wheel-0.45.1-py3-none-any.whl.metadata (2.3 kB)
Requirement already satisfied: rich in /usr/local/lib/python3.11/dist-packages (from keras>=3.5.0->tensorflow) (14.0.0)
Requirement already satisfied: namex in /usr/local/lib/python3.11/dist-packages (from keras>=3.5.0->tensorflow) (0.0.8)
Requirement already satisfied: optree in /usr/local/lib/python3.11/dist-packages (from keras>=3.5.0->tensorflow) (0.15.0)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from requests<3,>=2.21.0->tensorflo
```

```
import cv2
import os
import numpy as np
import tensorflow as tf
from tensorflow.keras.models import Model
from tensorflow.keras.layers import Input, TimeDistributed, Conv2D, MaxPooling2D, Flatten, Dense, LSTM, Dropout
from tensorflow.keras.applications import MobileNetV2
from tensorflow.keras.applications.mobilenet_v2 import preprocess_input
from tqdm import tqdm
```

Loading prepared video dataset (396 fake and 44 real videos)

```
!pip install google.colab
```

```
Requirement already satisfied: google.colab in /usr/local/lib/python3.11/dist-packages (1.0.0)
Requirement already satisfied: google-auth==2.38.0 in /usr/local/lib/python3.11/dist-packages (from google.colab) (2.38.0)
Requirement already satisfied: ipykernel==6.17.1 in /usr/local/lib/python3.11/dist-packages (from google.colab) (6.17.1)
Requirement already satisfied: ipyparallel==8.8.0 in /usr/local/lib/python3.11/dist-packages (from google.colab) (8.8.0)
Requirement already satisfied: ipython==7.34.0 in /usr/local/lib/python3.11/dist-packages (from google.colab) (7.34.0)
Requirement already satisfied: notebook==6.5.7 in /usr/local/lib/python3.11/dist-packages (from google.colab) (6.5.7)
Requirement already satisfied: pandas==2.2.2 in /usr/local/lib/python3.11/dist-packages (from google.colab) (2.2.2)
Requirement already satisfied: portpicker==1.5.2 in /usr/local/lib/python3.11/dist-packages (from google.colab) (1.5.2)
Requirement already satisfied: requests==2.32.3 in /usr/local/lib/python3.11/dist-packages (from google.colab) (2.32.3)
Requirement already satisfied: tornado==6.4.2 in /usr/local/lib/python3.11/dist-packages (from google.colab) (6.4.2)
```

```
Requirement already satisfied: cachetools<6.0,>=2.0.0 in /usr/local/lib/python3.11/dist-packages (from google-auth==2.38.0->google.cola
Requirement already satisfied: pyasn1-modules>=0.2.1 in /usr/local/lib/python3.11/dist-packages (from google-auth==2.38.0->google.colab
Requirement already satisfied: rsa<5,>=3.1.4 in /usr/local/lib/python3.11/dist-packages (from google-auth==2.38.0->google.colab) (4.9)
Requirement already satisfied: debugpy>=1.0 in /usr/local/lib/python3.11/dist-packages (from ipykernel==6.17.1->google.colab) (1.8.0)
Requirement already satisfied: jupyter-client>=6.1.12 in /usr/local/lib/python3.11/dist-packages (from ipykernel==6.17.1->google.colab)
Requirement already satisfied: matplotlib-inline>=0.1 in /usr/local/lib/python3.11/dist-packages (from ipykernel==6.17.1->google.colab)
Requirement already satisfied: nest-asyncio in /usr/local/lib/python3.11/dist-packages (from ipykernel==6.17.1->google.colab) (1.6.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.11/dist-packages (from ipykernel==6.17.1->google.colab) (24.2)
Requirement already satisfied: psutil in /usr/local/lib/python3.11/dist-packages (from ipykernel==6.17.1->google.colab) (5.9.5)
Requirement already satisfied: pyzmq>=17 in /usr/local/lib/python3.11/dist-packages (from ipykernel==6.17.1->google.colab) (24.0.1)
Requirement already satisfied: traitlets>=5.1.0 in /usr/local/lib/python3.11/dist-packages (from ipykernel==6.17.1->google.colab) (5.7.
Requirement already satisfied: decorator in /usr/local/lib/python3.11/dist-packages (from ipyparallel==8.8.0->google.colab) (5.2.1)
Requirement already satisfied: entrypoints in /usr/local/lib/python3.11/dist-packages (from ipyparallel==8.8.0->google.colab) (0.4)
Requirement already satisfied: python-dateutil>=2.1 in /usr/local/lib/python3.11/dist-packages (from ipyparallel==8.8.0->google.colab)
Requirement already satisfied: tqdm in /usr/local/lib/python3.11/dist-packages (from ipyparallel==8.8.0->google.colab) (4.67.1)
Requirement already satisfied: setuptools>=18.5 in /usr/local/lib/python3.11/dist-packages (from ipython==7.34.0->google.colab) (75.2.0
Collecting jedi>=0.16 (from ipython==7.34.0->google.colab)
  Downloading jedi-0.19.2-py2.py3-none-any.whl.metadata (22 kB)
Requirement already satisfied: pickleshare in /usr/local/lib/python3.11/dist-packages (from ipython==7.34.0->google.colab) (0.7.5)
Requirement already satisfied: prompt-toolkit!=3.0.0,!=3.0.1,<3.1.0,>=2.0.0 in /usr/local/lib/python3.11/dist-packages (from ipython==7
Requirement already satisfied: pygments in /usr/local/lib/python3.11/dist-packages (from ipython==7.34.0->google.colab) (2.19.1)
Requirement already satisfied: backcall in /usr/local/lib/python3.11/dist-packages (from ipython==7.34.0->google.colab) (0.2.0)
Requirement already satisfied: pexpect>4.3 in /usr/local/lib/python3.11/dist-packages (from ipython==7.34.0->google.colab) (4.9.0)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.11/dist-packages (from notebook==6.5.7->google.colab) (3.1.6)
Requirement already satisfied: argon2-cffi in /usr/local/lib/python3.11/dist-packages (from notebook==6.5.7->google.colab) (23.1.0)
Requirement already satisfied: jupyter-core>=4.6.1 in /usr/local/lib/python3.11/dist-packages (from notebook==6.5.7->google.colab) (5.7
Requirement already satisfied: ipython-genutils in /usr/local/lib/python3.11/dist-packages (from notebook==6.5.7->google.colab) (0.2.0)
Requirement already satisfied: nbformat in /usr/local/lib/python3.11/dist-packages (from notebook==6.5.7->google.colab) (5.10.4)
Requirement already satisfied: nbconvert>=5 in /usr/local/lib/python3.11/dist-packages (from notebook==6.5.7->google.colab) (7.16.6)
Requirement already satisfied: Send2Trash>=1.8.0 in /usr/local/lib/python3.11/dist-packages (from notebook==6.5.7->google.colab) (1.8.3
Requirement already satisfied: terminado>=0.8.3 in /usr/local/lib/python3.11/dist-packages (from notebook==6.5.7->google.colab) (0.18.1
Requirement already satisfied: prometheus-client in /usr/local/lib/python3.11/dist-packages (from notebook==6.5.7->google.colab) (0.21.
Requirement already satisfied: nbclassic>=0.4.7 in /usr/local/lib/python3.11/dist-packages (from notebook==6.5.7->google.colab) (1.2.0)
Requirement already satisfied: numpy>=1.23.2 in /usr/local/lib/python3.11/dist-packages (from pandas==2.2.2->google.colab) (2.0.2)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.11/dist-packages (from pandas==2.2.2->google.colab) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.11/dist-packages (from pandas==2.2.2->google.colab) (2025.2)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from requests==2.32.3->google.colab
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/dist-packages (from requests==2.32.3->google.colab) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/dist-packages (from requests==2.32.3->google.colab) (2.3
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.11/dist-packages (from requests==2.32.3->google.colab) (202
Requirement already satisfied: parso<0.9.0,>=0.8.4 in /usr/local/lib/python3.11/dist-packages (from jedi>=0.16->ipython==7.34.0->google
Requirement already satisfied: platformdirs>=2.5 in /usr/local/lib/python3.11/dist-packages (from jupyter-core>=4.6.1->notebook==6.5.7-
Requirement already satisfied: notebook-shim>=0.2.3 in /usr/local/lib/python3.11/dist-packages (from nbclassic>=0.4.7->notebook==6.5.7-
Requirement already satisfied: beautifulsoup4 in /usr/local/lib/python3.11/dist-packages (from nbconvert>=5->notebook==6.5.7->google.co
Requirement already satisfied: bleach!=5.0.0 in /usr/local/lib/python3.11/dist-packages (from bleach[css]!=5.0.0->nbconvert>=5->noteboo
Requirement already satisfied: defusedxml in /usr/local/lib/python3.11/dist-packages (from nbconvert>=5->notebook==6.5.7->google.colab)
Requirement already satisfied: jupyterlab-pygments in /usr/local/lib/python3.11/dist-packages (from nbconvert>=5->notebook==6.5.7->goog
```

```
from google.colab import drive
drive.mount("/content/drive")
base_path = "/content/drive/MyDrive/prepared_video_dataset"
```

```
Mounted at /content/drive
```

```
categories = ["Real", "Fake"]          # folders


# Video & image settings
num_frames = 10
height, width, channels = 224, 224, 3


# Containers
X = []
y = []



def extract_frames(video_path, num_frames, target_size) :
    cap = cv2.VideoCapture(video_path)
    total_frames = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))

    frame_idxs = np.linspace(0, total_frames - 1, num_frames).astype(int)
    frames = []

    for i in range(total_frames) :
        ret, frame = cap.read()
        if not ret :
            break
        if i in frame_idxs :
            frame = cv2.resize(frame, target_size)
            frame = frame.astype("float32") / 255.0
            frames.append(frame)
```

```
    cap.release()

    if len(frames) == num_frames :
        return np.array(frames)
    else:
        return None
```

```
for label, category in enumerate(categories) :  # Real = 0, Fake = 1
    folder_path = os.path.join(base_path, category)
    video_files = os.listdir(folder_path)

    for video_name in tqdm(video_files, desc=f"Processing {category}") :
        video_path = os.path.join(folder_path, video_name)
        frames = extract_frames(video_path, num_frames, (height, width))
        if frames is not None :
            X.append(frames)
            y.append(label)
```

```
Processing Real: 100%|██████████| 44/44 [00:02<00:00, 21.35it/s]
Processing Fake: 100%|██████████| 396/396 [00:19<00:00, 20.58it/s]
```

After dataset has been loaded

```
X = np.array(X)
y = np.array(y)
print(X)
print(y)
```

```
[[[[-0.9974164  -0.99723184 -0.99695504]
   [-0.9974164  -0.99723184 -0.99695504]
   [-0.9974164  -0.99723184 -0.99695504]
   ...
   [-0.9977547  -0.9976009  -0.9972626 ]
   [-0.9977547  -0.9976009  -0.9972626 ]
   [-0.9977547  -0.9976009  -0.9972626 ]]

  [[-0.9974164  -0.99723184 -0.99695504]
   [-0.9974164  -0.99723184 -0.99695504]
   [-0.9974164  -0.99723184 -0.99695504]
   ...
   [-0.9977547  -0.9976009  -0.9972626 ]
   [-0.9977547  -0.9976009  -0.9972626 ]
   [-0.9977547  -0.9976009  -0.9972626 ]]

  [[-0.9974164  -0.99723184 -0.99695504]
   [-0.9974164  -0.99723184 -0.99695504]
   [-0.9974164  -0.99723184 -0.99695504]
   ...
   [-0.9977547  -0.9976009  -0.9972626 ]
   [-0.9977547  -0.9976009  -0.9972626 ]
   [-0.9977547  -0.9976009  -0.9972626 ]]

  ...

  [[-0.9984929  -0.99858516 -0.9984006 ]
   [-0.9985544  -0.9986159  -0.9984314 ]
   [-0.99858516 -0.9986159  -0.9984929 ]
   ...
   [-0.99434066 -0.9945867  -0.9938485 ]
   [-0.9945867  -0.9948635  -0.99412537]
   [-0.994925   -0.9952018  -0.9944637 ]]

  [[-0.9983699  -0.9984314  -0.9982776 ]
   [-0.9983391  -0.9984006  -0.9982776 ]
   [-0.99830836 -0.9983699  -0.9982161 ]
   ...
   [-0.9944637  -0.9947405  -0.9939408 ]
   [-0.9947405  -0.9950173  -0.99427915]
   [-0.9950788  -0.9953556  -0.99461746]]

  [[-0.9980623  -0.9981238  -0.9979392 ]
   [-0.99800074 -0.9980623  -0.9979085 ]
   [-0.9979085  -0.99797    -0.9978162 ]
   ...
   [-0.9945252  -0.994802   -0.9940638 ]
   [-0.99483275 -0.99510956 -0.9943714 ]
   [-0.9952018  -0.9954787  -0.9947405 ]]]


 [[[-0.9974164  -0.99723184 -0.99695504]
   [-0.9974164  -0.99723184 -0.99695504]
   [-0.9974164  -0.99723184 -0.99695504]
   ...
   [-0.99772394 -0.99757016 -0.99723184]
```

```
[-0.99772394 -0.99757016 -0.99723184]
[-0.99772394 -0.99757016 -0.99723184]]
```

```
# Preprocess the data (you would replace this with your actual dataset and pipeline)
X = preprocess_input(X)
```

```
# Build the model
input_layer = Input(shape = (num_frames, height, width, channels))
```

```
# Base CNN model (MobileNetV2 without top)
base_input = Input(shape = (height, width, channels))
base_output = MobileNetV2(weights = "imagenet", include_top = False)(base_input)
base_model = Model(inputs = base_input, outputs = base_output)
base_model.trainable = False
```

```
<ipython-input-12-ea44f823bce8>:3: UserWarning: `input_shape` is undefined or non-square, or `rows` is not in [96, 128, 160, 192, 224]. W
  base_output = MobileNetV2(weights = "imagenet", include_top = False)(base_input)
Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/mobilenet_v2/mobilenet_v2_weights_tf_dim_ordering_tf_k
9406464/9406464 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
```

```
# Wrap CNN in TimeDistributed
cnn = TimeDistributed(base_model)(input_layer)
cnn = TimeDistributed(MaxPooling2D(pool_size = (2, 2)))(cnn)
cnn = TimeDistributed(Flatten())(cnn)
```

```
# Add LSTM layer
x = LSTM(64, implementation=1, unroll=True)(cnn)
x = Dropout(0.5)(x)
x = Dense(64, activation="relu")(x)
out = Dense(1, activation="sigmoid")(x)

model = Model(inputs=input_layer, outputs=out)
```

```
# Compile the model
model.compile(optimizer="adam", loss="binary_crossentropy", metrics=["accuracy"])
model.summary()
```

Model: "functional_1"

Layer (type)	Output Shape	Param #
input_layer (InputLayer)	(None, 10, 224, 224, 3)	0
time_distributed (TimeDistributed)	(None, 10, 7, 7, 1280)	2,257,984
time_distributed_1 (TimeDistributed)	(None, 10, 3, 3, 1280)	0
time_distributed_2 (TimeDistributed)	(None, 10, 11520)	0
lstm (LSTM)	(None, 64)	2,965,760
dropout (Dropout)	(None, 64)	0
dense (Dense)	(None, 64)	4,160
dense_1 (Dense)	(None, 1)	65

Total params: 5,227,969 (19.94 MB)
Trainable params: 2,969,985 (11.33 MB)

```
# Train the model (use a small number of epochs for testing)
model.fit(X, y, epochs = 5, batch_size = 4, validation_split = 0.2)
```

```
Epoch 1/5
88/88 ━━━━━━━━━━━━━━━━━━━━ 319s 3s/step - accuracy: 0.8694 - loss: 0.3993 - val_accuracy: 1.0000 - val_loss: 0.1624
Epoch 2/5
88/88 ━━━━━━━━━━━━━━━━━━━━ 270s 3s/step - accuracy: 0.8811 - loss: 0.4039 - val_accuracy: 1.0000 - val_loss: 0.1446
Epoch 3/5
88/88 ━━━━━━━━━━━━━━━━━━━━ 272s 3s/step - accuracy: 0.8946 - loss: 0.3666 - val_accuracy: 1.0000 - val_loss: 0.1771
Epoch 4/5
88/88 ━━━━━━━━━━━━━━━━━━━━ 271s 3s/step - accuracy: 0.8813 - loss: 0.3539 - val_accuracy: 1.0000 - val_loss: 0.1514
Epoch 5/5
88/88 ━━━━━━━━━━━━━━━━━━━━ 269s 3s/step - accuracy: 0.8677 - loss: 0.4128 - val_accuracy: 1.0000 - val_loss: 0.1308
<keras.src.callbacks.history.History at 0x7875842dcfd0>
```

```
# Convert to TFLite
converter = tf.lite.TFLiteConverter.from_keras_model(model)
```

```
tflite_model = converter.convert()

# Save the TFLite model
with open("deepfake_detection_model.tflite", "wb") as f:
    f.write(tflite_model)
    print("TFLite model saved.")
```

```
Saved artifact at '/tmp/tmpdonvzrv8'. The following endpoints are available:

  * Endpoint 'serve'
    args_0 (POSITIONAL_ONLY): TensorSpec(shape=(None, 10, 224, 224, 3), dtype=tf.float32, name='keras_tensor')
  Output Type:
    TensorSpec(shape=(None, 1), dtype=tf.float32, name=None)
  Captures:
    132446169963344: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446169964112: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446169966800: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446169966416: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446169965264: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446169966992: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446169966032: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446169963536: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446169967184: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446169965072: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446169968528: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446169970640: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446169970064: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446169970448: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446169969104: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446169969680: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446169969488: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125588560: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446169972560: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446169972368: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125590288: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125589904: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125589328: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125590480: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125588944: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125592592: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125592976: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125593360: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125593168: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125588752: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125594512: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125594896: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125595280: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125595088: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125591440: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125596432: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125596816: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125597200: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125597008: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125592208: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125594128: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125598928: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125599312: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125599120: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125595664: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125597968: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125601040: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125601424: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125601232: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125598352: TensorSpec(shape=(), dtype=tf.resource, name=None)
    132446125602576: TensorSpec(shape=(), dtype=tf.resource, name=None)
```

Evaluation metrics

```
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
from sklearn.metrics import classification_report, confusion_matrix, ConfusionMatrixDisplay

# Get predictions
y_pred_probs = model.predict(X)
y_pred = (y_pred_probs > 0.5).astype(int)
```

```
14/14 ━━━━━━━━━━━━━━━━━━━━ 225s 16s/step
```

```
# Print metrics
print("Accuracy :", accuracy_score(y, y_pred))
print("Precision :", precision_score(y, y_pred))
print("Recall :", recall_score(y, y_pred))
print("F1 Score :", f1_score(y, y_pred))
print("Confusion Matrix :\n", confusion_matrix(y, y_pred))
```

```
Accuracy : 0.9
Precision : 0.9
Recall : 1.0
F1 Score : 0.9473684210526315
Confusion Matrix :
 [[  0  44]
 [  0 396]]
```

```
disp = ConfusionMatrixDisplay(confusion_matrix(y, y_pred), display_labels=["Real", "Fake"])
disp.plot()
```

Show hidden output